

Functions $[x]$ and $\{x\}$

Definitions and properties

In dealing with many questions, it is often useful to separate a real number x into two parts, namely its *integral part*, $[x]$, which is the largest integer less than or equals to x , and the remaining part $x - [x]$, which is the *fractional part* of x , and is denoted by $\{x\}$. Thus $x = [x] + \{x\}$, with $0 \leq \{x\} < 1$. The function $y = [x]$ is called the *integer function*. It was first used by Guass, thus it is also called the *Gaussian function*. It is easy to see that the graph of the Gaussian function consists of horizontal segments of length 1, starting from an integer n to real x less than $n+1$. (Examples: $[5] = 5$; $[3.1416] = 3$ and $\{3.1416\} = 0.1416$; $[\pi] = 3$ and its fractional part $\{\pi\}$ lies between 0.1415 and 0.1416; $[-7.82] = -8$ and $\{-7.82\} = 0.18$.)

In combinatorics other notations are used. Namely the largest integer less than or equals to x is called the *Floor* of x , and is denoted by $\lfloor x \rfloor$. On the other hand, the smallest integer larger than or equals to x is called the *Ceiling* of x , and is denoted by $\lceil x \rceil$. Hence $\lfloor x \rfloor = \lceil x \rceil$ if and only if x is an integer.

We list basic *properties* of the Gaussian function.

1. $x - 1 < [x] \leq x < [x] + 1$.
2. If $x \in [0, 1)$, i.e. x is larger than or equals to 0 and is strictly less than 1, then $[x] = 0$.
3. $y = [x]$ is a non-decreasing function, i.e. if $x_1 \leq x_2$, then $[x_1] \leq [x_2]$.
4. If x is an integer, then $[-x] = -[x]$; if x is *not* an integer, then $[-x] = -[x] - 1$.
5. For any real number x , $[[x]] = [x]$.
6. For any real number x and any integer n , $[n + x] = n + [x]$.

Proof: Indeed $x - 1 < [x] \leq x$, hence $n + x - 1 < n + [x] \leq n + x$. This means $n + [x]$ is an integer within the interval $(n + x - 1, n + x]$, and so is $[n + x]$. But there is only one integer within the interval, thus $[n + x] = n + [x]$.

7. If $x, y \geq 0$, then $[xy] \geq [x][y]$.

Proof: Since $x \geq [x] \geq 0, y \geq [y] \geq 0$, we have $xy \geq [x][y]$. By the non-decreasing property of the integer function, the result follows.

8. If $[x] = [y]$, then $|x - y| < 1$.

Proof: Let $x = [x] + \{x\}$ and $y = [y] + \{y\}$. Then $|x - y| = |\{x\} - \{y\}| < 1$.

9. Let n be a natural number, then $[nx] \geq n[x]$.

Proof: Check that it suffices to show the statement is true when $0 \leq x < 1$. But in this case it is obvious, with equality holds when $x = 0$. In general the equality holds when x is an integer.

10. $[x] + [y] \leq [x + y] \leq [x] + [y] + 1$.

Proof: It is equivalent to show $[x] + [y] \leq [[x] + [y] + \{x\} + \{y\}] \leq [x] + [y] + 1$. The middle term equals to $[x] + [y] + [\{x\} + \{y\}]$. As $0 \leq \{x\} + \{y\} < 2$, $0 \leq [\{x\} + \{y\}] \leq 1$, and the result follows.

11. $[x] - [y] - 1 \leq [x - y] \leq [x] - [y]$.

12. If n is a natural number, then $\left[\frac{[x]}{n} \right] = \left[\frac{x}{n} \right]$.

Proof: $\frac{x}{n} = \left[\frac{x}{n} \right] + \left\{ \frac{x}{n} \right\}$, hence $x = n \left[\frac{x}{n} \right] + n \left\{ \frac{x}{n} \right\}$ and $[x] = n \left[\frac{x}{n} \right] + [n \left\{ \frac{x}{n} \right\}]$. Therefore

$$\left[\frac{[x]}{n} \right] = \left[\frac{x}{n} \right] + \left[\frac{n \left\{ \frac{x}{n} \right\}}{n} \right]. \text{ But } 0 \leq n \left\{ \frac{x}{n} \right\} < n, \text{ hence } \left[\frac{n \left\{ \frac{x}{n} \right\}}{n} \right] = 0. \text{ The result follows.}$$

13. When an integer a is divided by a natural number b , its quotient is $\left[\frac{a}{b} \right]$ and the remainder $a - \left[\frac{a}{b} \right] b$.

14. The highest power of a prime number p that divides $n!$ is $\left[\frac{n}{p} \right] + \left[\frac{n}{p^2} \right] + \left[\frac{n}{p^3} \right] + \dots$

This number is denoted by $v_p(n)$, the p -adic valuation of n at p .

Proof: Note that the sum is a finite sum because pretty soon (r large enough) we will

have $\left[\frac{n}{p^r} \right] = 0$. In fact among the integers $1, 2, \dots, n$, multiples of p are

$p, 2p, \dots, \left[\frac{n}{p} \right] p$, altogether $\left[\frac{n}{p} \right]$ of them. The multiples of p^2 are

$p^2, 2p^2, \dots, \left[\frac{n}{p^2} \right] p^2$, altogether $\left[\frac{n}{p^2} \right]$ of them, and so on. Hence $v_p(n) = \sum_{i \geq 1} \left[\frac{n}{p^i} \right]$.

Suppose now $n = n_l p^l + n_{l-1} p^{l-1} + \dots + n_1 p + n_0$ (p -adic expansion of n). Then it is easy

To check $\left[\frac{n}{p} \right] = n_l p^{l-1} + n_{l-1} p^{l-2} + \dots + n_1$, and $\left[\frac{n}{p^2} \right] = n_l p^{l-2} + n_{l-1} p^{l-3} + \dots + n_2$, and so

on. Summing up all terms with the same coefficients (n_i), one can show

$$v_p(n) = \sum_{i \geq 1} \left[\frac{n}{p^i} \right] = \frac{n - s(n)}{p - 1},$$

where $s(n) = n_l + n_{l-1} + \dots + n_1 + n_0$, the sum of coefficients of the p -adic expansion of n .

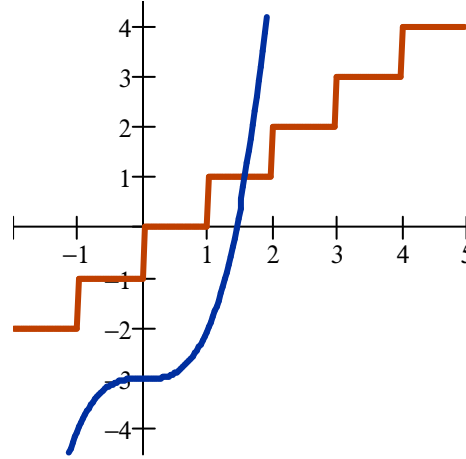
Finally the formula is called the *Legendre's formula*.

Equations and inequalities

When dealing with equations or inequalities involving $[x]$ or $\{x\}$, we either use the basic definitions, resolve them to other forms using basic inequalities or identities, separate the problem into suitable regions, or occasionally we can also use graphical methods.

Example 1: Solve the equation $x^3 - [x] = 3$.

The equation is equivalent to $-x^3 + x + 3 = \{x\}$, hence $0 \leq -x^3 + x + 3 < 1$, get $3 \geq x(x^2 - 1) > 2$. One must have $x > 0$, otherwise $x^2 - 1 < 0$, or $0 \leq |x| < 1$, hence $|x^2 - 1| \leq 1$ and one cannot have $|x(x^2 - 1)| \geq 2$. On the other hand, $3 \geq x(x^2 - 1)$, hence $x < 2$. Thus $[x] = 0, 1$. The first option produces no solution and the second one implies $x = \sqrt[3]{4}$. (The graphs of $[x]$ and $x^3 - 3$ indeed meet at one point.)



Example 2: $[2x] + [2y] \geq [x] + [x + y] + [y]$.

First note that for $\alpha, \beta \in [0, 1)$,

$[2\alpha] + [2\beta] \geq [\alpha + \beta]$. For $[\alpha + \beta] = 0$ or 1, in the first case there is nothing to prove, in the second case at least one of α, β equals 0.5 or more, hence one of $[2\alpha]$ or $[2\beta]$ equals 1. Now

$$\begin{aligned} [2x] + [2y] &= [2[x] + 2\{x\}] + [2[y] + 2\{y\}] \\ &\geq [x] + [y] + [[x] + [y] + \{x\} + \{y\}] \\ &= [x] + [y] + [x + y]. \end{aligned}$$

Example 3: Find the sum of roots of $[3x + 1] = 2x - \frac{1}{2}$.

Indeed $2x - \frac{1}{2} = t$ is an integer, hence $x = \frac{1}{2}t + \frac{1}{4}$. From $0 \leq (3x + 1) - (2x - \frac{1}{2}) < 1$, get $0 \leq (\frac{3}{2}t + \frac{7}{4}) - t < 1$, or $t = -2, -3$, hence $x = -\frac{3}{4}, -\frac{5}{4}$. The sum is -2 .

Example 4: Solve $[x^2] - [x] - 2 = 0$.

As $[x^2] = [x] + 2$, we have $[x] + 2 \geq 0$. Moreover $[x] \leq 2$, otherwise $[x] \geq 3$, get $[x^2] - [x] \geq [x]([x] - 1) > 6$, a contradiction. Hence $[x] = -2, -1, 0, 1, 2$. From these, get possible region of x and the solutions $x = -1$ or $x \in [\sqrt{3}, \sqrt{5})$.

Example 5 (RMC 2004): (a) Find infinitely many rational positive numbers x such that $\{x^2\} + \{x\} = 0.99$. (b) Can you find rational positive x such that $\{x^2\} + \{x\} = 1$?

(a) As $0.99 = \frac{99}{100}$, we look for numbers of the form $\frac{n}{10}$, and we find $x = \frac{13}{10}$ works, hence all numbers of the form $10k + \frac{13}{10}$ work. (b) No! If x is such a number and $[x] = n$, then $n < x < n+1$, hence $n^2 < x^2 < n^2 + 2n + 1$. Therefore $[x^2] = n^2 + k$, with $0 \leq k \leq 2n$. Put all these into the equation, get $x^2 - n^2 - k + x - n = 1$. Now x is rational, hence the discriminant $4n^2 + 4n + 4k + 5$ is an odd square $(2m+1)^2$, hence $n^2 + n + k + 1 = m^2 + m$. This forces $m \geq n+1$, hence $m^2 + m \geq n^2 + 3n + 2$. Together we have $k \geq 2n+1$, contradicting the condition $0 \leq k \leq 2n$.

Calculating the values of certain expressions or verifications of identities

Example 7: Find the day of the week of A.D. m year, n month and p days.

Indeed A.D. 1 January 1st is a Monday. From this day to A.D. m years, n months and p days altogether A days have gone. The day of the week is $x = A - 7 \left\lfloor \frac{A}{7} \right\rfloor$. Now every four year is a leap year, but every 100 years, the supposed leap year is not counted, but every 400 years the leap year is counted again. Hence

$A = 365(m-1) + \left\lfloor \frac{m-1}{4} \right\rfloor - \left\lfloor \frac{m-1}{100} \right\rfloor + \left\lfloor \frac{m-1}{400} \right\rfloor + T$, where T is the number of days of the m year from January 1st to n months and p days. Hence

$x = B + T - 7 \left\lfloor \frac{B+T}{7} \right\rfloor$, where $B = m-1 + \left\lfloor \frac{m-1}{4} \right\rfloor - \left\lfloor \frac{m-1}{100} \right\rfloor + \left\lfloor \frac{m-1}{400} \right\rfloor$.

Today is August 14, 2007. From January 1st to this day there are altogether 226 days.

Also $B = 2007 - 1 + \left\lfloor \frac{2007-1}{4} \right\rfloor - \left\lfloor \frac{2007-1}{100} \right\rfloor + \left\lfloor \frac{2007-1}{400} \right\rfloor = 2492$. Hence

$x = 2492 + 226 - 7 \left\lfloor \frac{2492 + 226}{7} \right\rfloor = 2$, i.e. today is a Tuesday.

Example 8: Determine the number of different values in the sequence

$$\left\lfloor \frac{1^2}{2007} \right\rfloor, \left\lfloor \frac{2^2}{2007} \right\rfloor, \dots, \left\lfloor \frac{2007^2}{2007} \right\rfloor.$$

If $x - y > 1$, then $[x]$ and $[y]$ will differ. Now $\frac{(k+1)^2}{2007} - \frac{k^2}{2007} > 1$ when $k > 1003$. Hence from $k = 1004$ to $k = 2007$ (1004 terms), all terms will differ. Now $\left\lceil \frac{1003^2}{2007} \right\rceil = 501$ and $\left\lceil \frac{1004^2}{2007} \right\rceil = 502$. All integer values from 0 to 501 will appear. Hence altogether there are $502 + 1004 = 1506$ different values.

Example 9: Let $S = [\sqrt{1}] + [\sqrt{2}] + \dots + [\sqrt{2007}]$. Find $[\sqrt{S}]$.

Consider $T = ([\sqrt{1}] + [\sqrt{2}] + [\sqrt{3}]) + ([\sqrt{4}] + \dots + [\sqrt{8}]) + \dots + ([\sqrt{(n-1)^2}] + \dots + [\sqrt{n^2 - 1}])$. In the i^{th} group, there are $((i+1)^2 - 1) - i^2 + 1 = 2i + 1$ terms, each with value i , hence $T = \sum_{i=1}^{n-1} i(2i+1) = 2 \sum_{i=1}^{n-1} i^2 + \sum_{i=1}^{n-1} i = \frac{1}{6}n(n-1)(4n+1)$. Take $n = 44$, get $T = \frac{1}{6} \times 44 \times 43 \times 177$.

Also $2007 - 1936 + 1 = 72$, hence $S = \frac{1}{6} \times 44 \times 43 \times 177 + 44 \times 72 = 58982$, and $[\sqrt{S}] = 242$.

Example 10 (IMO 1968): For any natural number n , evaluate $\sum_{k=0}^{\infty} \left\lceil \frac{n+2^k}{2^{k+1}} \right\rceil$.

Indeed after some experimentation it is not hard to see that the (finite) sum equals n . Moreover it is natural to use the binary expansion of n . Let $n = a_k 2^k + a_{k-1} 2^{k-1} + \dots + a_0$,

with $a_k = 1$ and $a_j = 0$ or 1 , $0 \leq j \leq k-1$. Then $\left\lceil \frac{n+2^j}{2^{j+1}} \right\rceil = 0$ for $j \geq k+1$. For any j ,

$0 \leq j \leq k-1$, we have $\frac{n+2^j}{2^{j+1}} = 2^{k-j-1} a_k + \dots + a_{j+1} + \frac{2^j + 2^j a_j + \dots + 2a_1 + a_0}{2^{j+1}}$. Check that if

$a_j = 0$, then $\left\lceil \frac{2^j + 2^j a_j + \dots + 2a_1 + a_0}{2^{j+1}} \right\rceil = 0$, and if $a_j = 1$, then

$\left\lceil \frac{2^j + 2^j a_j + \dots + 2a_1 + a_0}{2^{j+1}} \right\rceil = 1$. Hence $\left\lceil \frac{n+2^j}{2^{j+1}} \right\rceil = 2^{k-j-1} a_k + \dots + a_{j+1} + a_j$ exactly. The

required sum equals

$$\begin{aligned} \sum_{j=0}^k \left\lceil \frac{n+2^j}{2^{j+1}} \right\rceil &= (2^{k-1} a_k + 2^{k-2} a_{k-1} + \dots + 2a_2 + a_1 + a_0) \\ &+ (2^{k-2} a_k + \dots + a_2 + a_1) + \dots + (2a_k + a_{k-1} + a_{k-1}) + a_k + a_k \\ &= a_k (2^{k-1} + 2^{k-2} + \dots + 2 + 1 + 1) + a_{k-1} (2^{k-2} + \dots + 2 + 1 + 1) + \dots + (a_1 + a_1) + a_0 \\ &= a_k 2^k + a_{k-1} 2^{k-1} + \dots + 2a_1 + a_0 = n. \end{aligned}$$

Alternative Solution: Check that $\left\lfloor x + \frac{1}{2} \right\rfloor = [2x] - [x]$. This can be verified by considering

the cases $k \leq x < k + \frac{1}{2}$ and $k + \frac{1}{2} \leq x < k + 1$. Now

$$\sum_{j=0}^k \left\lfloor \frac{n+2^j}{2^{j+1}} \right\rfloor = \sum_{j=0}^k \left\lfloor \frac{n}{2^{j+1}} + \frac{1}{2} \right\rfloor = \sum_{j=0}^k \left(\left\lfloor \frac{n}{2^j} \right\rfloor - \left\lfloor \frac{n}{2^{j+1}} \right\rfloor \right) = n.$$

Example 11 (Hermite's Identity): Show that for any natural number n and real number x , $[x] + [x + \frac{1}{n}] + [x + \frac{2}{n}] + \dots + [x + \frac{n-1}{n}] = [nx]$.

Put $x = [x] + \alpha, \alpha \in [0, 1)$ into the equation and by eliminating the common terms, the equation becomes $[\alpha] + [\alpha + \frac{1}{n}] + [\alpha + \frac{2}{n}] + \dots + [\alpha + \frac{n-1}{n}] = [n\alpha]$. Hence it suffices to consider only those $x \in (0, 1)$. Now suppose $x \in [\frac{k-1}{n}, \frac{k}{n})$, i.e. x belongs to the k^{th} interval, then $k-1 \leq nx < k$, hence $[nx] = k-1$. On the other hand, only (and only) each of the terms $[x + \frac{n-k+1}{n}], [x + \frac{n-k+2}{n}], \dots, [x + \frac{n-1}{n}]$ ($k-1$ of them) contributes one "1" to the left hand side, and the result follows.

Example 12 (USAMO 1981): Let x be a positive real number and n a natural number, then $[nx] \geq \frac{[x]}{1} + \frac{[2x]}{2} + \dots + \frac{[nx]}{n}$.

Let $x_n = \frac{[x]}{1} + \frac{[2x]}{2} + \dots + \frac{[nx]}{n}$. We proceed to show by induction that $[nx] \geq x_n$. The statement is true for $n=1$. Now let $[kx] \geq x_k$ for $k=1, 2, \dots, n-1$. Then $x_n = x_{n-1} + \frac{[nx]}{n}$.

Hence

$$nx_n = (n-1)x_{n-1} + x_{n-1} + [nx],$$

$$(n-1)x_{n-1} = (n-2)x_{n-2} + x_{n-2} + [(n-1)x],$$

...

$$3x_3 = 2x_2 + x_2 + [3x],$$

$$2x_2 = x_1 + x_1 + [2x].$$

By summing up the equations and canceling out the common terms, get

$$\begin{aligned} nx_n &= \sum_{i=1}^{n-1} x_i + x_i + \sum_{i=2}^n [ix] \leq \sum_{i=1}^{n-1} [ix] + [x] + \sum_{i=2}^n [ix] = 2 \sum_{i=1}^{n-1} [ix] + [nx] \\ &= (([x] + [(n-1)x]) + ([2x] + [(n-2)x]) + \dots + ([(n-1)x] + [x])) + [nx] \\ &\leq \underbrace{[nx] + \dots + [nx]}_{n-1 \text{ terms}} + [nx] = n[nx], \end{aligned}$$

the result follows.

More Number Theoretic Problems

Example 13 (IMO 1972): Let m and n be non-negative integers, show that $\frac{(2m)!(2n)!}{m!n!(m+n)!}$ is an integer.

In example 2 we have shown $[2x] + [2y] \geq [x] + [x+y] + [y]$. Now let p be a prime number, we have $\left\lfloor \frac{2m}{p^k} \right\rfloor + \left\lfloor \frac{2n}{p^k} \right\rfloor \geq \left\lfloor \frac{m}{p^k} \right\rfloor + \left\lfloor \frac{m+n}{p^k} \right\rfloor + \left\lfloor \frac{n}{p^k} \right\rfloor$. Summing k from 1 to ∞ we see that there are more p in the numerator of $\frac{(2m)!(2n)!}{m!n!(m+n)!}$ than in the denominator, hence the result.

Note: By using $[5x] + [5y] \geq [3x+y] + [3y+x]$, one can show $\frac{(5m)!(5n)!}{m!n!(3m+n)!(3n+m)!}$ is an integer (USAMO 1975).

Example 14 (Canada MO 1985): Show that 2^{n-1} divides $n!$ if and only if $n = 2^k$ for some natural number k .

If $n = 2^k$, then $\left\lfloor \frac{2^k}{2} \right\rfloor + \left\lfloor \frac{2^k}{2^2} \right\rfloor + \left\lfloor \frac{2^k}{2^3} \right\rfloor + \dots = 2^{k-1} + 2^{k-2} + \dots + 1 = 2^k - 1 = n - 1$, hence $2^{n-1} \mid n!$.

Conversely if $2^{n-1} \mid n!$, suppose $2^k \leq n < 2^{k+1}$, and $n = 2^k + a_{k-1}2^{k-1} + \dots + a_0$ be the binary expansion of n , with $a_j = 0$ or 1 , $0 \leq j \leq k-1$. One checks $\left\lfloor \frac{n}{2^r} \right\rfloor = 0$ if $r \geq k+1$,

$\left\lfloor \frac{n}{2^k} \right\rfloor = 1$ and $\left\lfloor \frac{n}{2^s} \right\rfloor = 2^{k-s} + a_{k-1}2^{k-s-1} + \dots + a_s$ for $1 \leq s \leq k-1$. Adding all these to get $\sum_{s=1}^{\infty} \left\lfloor \frac{n}{2^s} \right\rfloor = \sum_{s=1}^k \left\lfloor \frac{n}{2^s} \right\rfloor = n - 1 - \sum_{s=0}^{k-1} a_s$, and the expression equals $n-1$ only when $a_s = 0$, $0 \leq s \leq k-1$.

Example 15: Let p be an odd prime number, show that p divides $\left[(2 + \sqrt{5})^p \right] - 2^{p+1}$.

As $-1 < (2 - \sqrt{5})^p < 0$, hence $\left[(2 + \sqrt{5})^p \right] = (2 + \sqrt{5})^p + (2 - \sqrt{5})^p$. Use the binomial expansion to show that

$\left[(2 + \sqrt{5})^p \right] - 2^{p+1} = 2(C_2^p \cdot 2^{p-2} \cdot 5 + C_4^p \cdot 2^{p-4} \cdot 5^2 + \dots + C_{p-1}^p \cdot 2^1 \cdot 5^{\frac{p-1}{2}})$, but each of the coefficient $C_2^p, C_4^p, \dots, C_{p-1}^p$ is divisible by p .

Example 16: In the decimal expansion of $(\sqrt{2} + \sqrt{3})^{2008}$, find the values of its digits just above and below its decimal place.

First $(\sqrt{2} + \sqrt{3})^{2008} = (5 + 2\sqrt{6})^{1004}$. Now note that $0 < (5 - 2\sqrt{6})^{1004} \approx 0.101^{1004} < 1$ and is real small. Then we have the integer

$$\begin{aligned} n &= (5 + 2\sqrt{6})^{1004} + (5 - 2\sqrt{6})^{1004} = 2(5^{2004} + C_2^{1004} \cdot 5^{1002} \cdot (2\sqrt{6})^2 + \dots + (2\sqrt{6})^{1004}) \\ &\equiv 2(24)^{502} \equiv 2 \cdot 6 \equiv 2 \pmod{10}. \end{aligned}$$

Hence $n = **...2$. Therefore $(5 + 2\sqrt{6})^{1004} = **...2 - 0.00...*** = **...1.99...**. Thus the digits are 1 and 9 respectively.$

Example 17 (Korea MO 2000): Let p be a prime such that $p \equiv 1 \pmod{4}$. Evaluate

$$\sum_{k=1}^{p-1} \left(\left\lfloor \frac{2k^2}{p} \right\rfloor - 2 \left\lfloor \frac{k^2}{p} \right\rfloor \right).$$

Using the relation $[x] = x - \{x\}$, the expression becomes $\sum_{k=1}^{p-1} \left(2 \left\{ \frac{k^2}{p} \right\} - \left\{ \frac{2k^2}{p} \right\} \right)$. Now if

$\{x\} < \frac{1}{2}$, $2\{x\} - \{2x\} = 0$. When $\{x\} \geq \frac{1}{2}$, $2\{x\} - \{2x\} = 2\{x\} - (2\{x\} - 1) = 1$. Therefore we

need to look for those $\left\{ \frac{k^2}{p} \right\} \geq \frac{1}{2}$, or equivalently, the number of nonzero residues k

modulo p such that k^2 is congruent to some number in $\left[\frac{p+1}{2}, p-1 \right]$ modulo p . Because

$p \equiv 1 \pmod{4}$, we know there exists d such that $d^2 \equiv -1 \pmod{p}$. Partition the nonzero residues into pairs $\{a, da\}$, then $a^2 \equiv -(da)^2 \pmod{p}$, we know exactly one in each pair has a square congruent to some number in $\left[\frac{p+1}{2}, p-1 \right]$, for a total of $\frac{p-1}{2}$ residues.

Generating subsets of integers

Some subsets of integers may be enumerated using certain integer functions.

Example 18: Show that the function $f(n) = [n + \sqrt{n} + \frac{1}{2}]$, $n \in N$, misses exactly the squares.

We need to consider those n such that $f(n+1) - f(n) \geq 2$ only, which is equivalent to find an integer k such that $\sqrt{n} + \frac{1}{2} < k \leq \sqrt{n+1} + \frac{1}{2}$. The inequality can be transformed to

$n < (k - \frac{1}{2})^2 = k^2 - k + \frac{1}{4} \leq n+1$. This forces $n = k^2 - k$. Thus from $f(n)$ to $f(n+1)$,

exactly the number $n+k=k^2$ is omitted. (Note exactly one number is missed, as $(\sqrt{n+1}+\frac{1}{2})-(\sqrt{n}+\frac{1}{2})<1$. Two sequences whose union is N but with empty intersection are *Complementary sequences*. In this example $[n+\sqrt{n}+\frac{1}{2}]$ and the squares are complementary.)

Example 19: Show that the function $f(n)=[n+\sqrt{\frac{n}{3}}+\frac{1}{2}]$ misses exactly the numbers of the form $3n^2-2n$.

Again consider those n such that $f(n+1)-f(n)\geq 2$, namely $\sqrt{\frac{n}{3}}+\frac{1}{2}<k\leq\sqrt{\frac{n+1}{3}}+\frac{1}{2}$, which can be transformed to $n<3(k-\frac{1}{2})^2\leq n+1$. This implies $n=3k^2-3k$. Thus from $f(n)$ to $f(n+1)$ exactly the number $n+k=3k^2-2k$ is missed.

Beatty's Theorem

An interesting property concerning integer functions is the following *Beatty's Theorem*.

Example 20: Let α and β be irrational numbers satisfying $\frac{1}{\alpha}+\frac{1}{\beta}=1$. Then the sequences $[n\alpha]$ and $[n\beta], n=1,2,\dots$ are disjoint and their union is N , the set of natural numbers.

Clearly $\alpha, \beta \in (1,2)$. Further $[n\alpha]$ and $[n\beta]$ are strictly increasing. Now let c be a natural number, there are h terms of $[n\alpha]$ in $[1,c)$ and there are k terms of $[n\beta]$ in $[1,c)$.

Then $[h\alpha]<c\leq[(h+1)\alpha]$, hence $h\alpha<c<(h+1)\alpha$ or $h<\frac{c}{\alpha}<h+1$, (note that α is

irrational. Likewise $k<\frac{c}{\beta}<k+1$. Adding up the two inequalities, get $h+k<c<h+k+2$. Thus $c=h+k+1$. Therefore in the range $[1,c)$, there are altogether $h+k=c-1$ terms of $[n\alpha]$ and $[n\beta]$. As c is arbitrary, it implies that there is exactly one term of $[n\alpha]$ and $[n\beta]$ in $[c,c+1)$. This term must equal c .

(Alternative proof): First show the two sequences are disjoint. Indeed if $[m\alpha]=[n\beta]=q$, then $q<m\alpha<q+1$ and $q<n\beta<q+1$ (α, β irrational). Hence $\frac{m}{q+1}<\frac{1}{\alpha}<\frac{m}{q}$ and

$\frac{n}{q+1} < \frac{1}{\beta} < \frac{n}{q}$. By adding them, get $q < m+n < q+1$, impossible. Now suppose $[q, q+1)$ contains no elements of $[m\alpha]$ and $[n\beta]$, then there exist m and n such that $m\alpha < q < q+1 < (m+1)\alpha$ and $n\beta < q < q+1 < (n+1)\beta$. Hence $\frac{m}{q} < \frac{1}{\alpha} < \frac{m+1}{q+1}$ and $\frac{n}{q} < \frac{1}{\beta} < \frac{n+1}{q+1}$. Adding up and simplify to get $m+n < q < q+1 < m+n+2$, again a contradiction.

An interesting application is found in the following example.

Example 21 (IMO 1978): The set of positive integers $N = \{1, 2, 3, \dots\}$ is the union of two disjoint subsets $\{f(1), f(2), \dots\}$ and $\{g(1), g(2), \dots\}$, where $f(1) < f(2) < f(3) < \dots$, $g(1) < g(2) < g(3) < \dots$ and $g(n) = f(f(n)) + 1$ for all $n \geq 1$. Determine $f(240)$.

We try to get several values first, indeed $g(1) = f(f(1)) + 1 > 1$, hence $f(1) = 1$ and $g(1) = f(f(1)) + 1 = 2$. Thus $f(2) \geq 3$ and $f(3) > f(2) \geq 3$, so $f(3) \geq 4$. Now $g(2) = f(f(2)) + 1 \geq f(3) + 1 \geq 5$. Therefore $f(2) = 3$, $f(3) = 4$ and hence $g(2) = 5$. Now we must have $f(4) \geq 6$, thus $g(3) = f(f(3)) + 1 \geq 7$. To fill the gap, we must have $f(4) = 6$ and $g(3) = 7$. Going further we have $f(5) \geq 8$, $f(6) \geq 9$, $g(4) = f(f(4)) + 1 \geq 10$. Therefore we must have $f(5) = 8$, $f(6) = 9$ and $g(4) = 10$, etc.

We now observe that $g(n) = f(f(n)) + 1$ is the n^{th} gap that separate the $f(n)$'s. The number $f(f(n))$ belongs to $\{f(n)\}$, it has $n-1$ gaps ($g(1)$ to $g(n-1)$) before it and it is the $f(n)^{\text{th}}$ member of $f(n)$'s. Therefore $f(f(n)) = f(n) + n - 1$, hence $g(n) = f(n) + n$. (*) Moreover because $g(n) - 1 = f(f(n))$, hence $g(n+1) - g(n) = f(f(n+1)) - f(f(n)) \geq 2$. Therefore $f(f(n)+1) = f(n) + n + 1$. (**) By (*) and (**) it is not hard to find $f(240) = 388$.

(Alternative solution) We try $f(n) = [\alpha n]$ and $g(n) = [\beta n]$, where α and β are irrational with $\frac{1}{\alpha} + \frac{1}{\beta} = 1$. The condition implies $[\beta n] = [\alpha[\alpha n]] + 1$. Dividing by n and

let n goes to infinity, get $\beta = \alpha^2$. Solve to get $\alpha = \frac{\sqrt{5}+1}{2}$ and $\beta = \frac{\sqrt{5}+3}{2}$. Now observe

$$\begin{aligned} [\alpha[\alpha n]] + 1 &= [\alpha^2 n - \alpha\{\alpha n\}] + 1 = [(\alpha+1)n - \alpha\{\alpha n\}] + 1 \\ &= n + [\alpha n] + [1 - (\alpha-1)\{\alpha n\}] = n + [\alpha n] = [(\alpha+1)n] = [\alpha^2 n] = [\beta n]. \end{aligned}$$

We give a table of the numbers.

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
$f(n)$	1	3	4	6	8	9	11	12	14	16	17	19	21	22	24	25	27
$g(n)$	2	5	7	10	13	15	18	20	23	26	28	31	34	36	39	41	44

Example 22 (Wythoff's Game): There are two piles of stones. Two persons take turn to remove several stones from the two piles. The rules are as follows. One may take away one or more (or all) the stones from one pile or an equal number of stones from each pile (at least one stone from each pile). The person who takes the last stone wins. Who has the winning strategy?

We define the numbers $(a_n, b_n) = (f(n), g(n))$ together with $(a_0, b_0) = (0, 0)$ and call them the *singular states*. Clearly the a_n in (a_n, b_n) is the smallest natural number which did not appear in (a_i, b_i) with $0 \leq i \leq n-1$. Moreover $b_n = a_n + n$. The distributions which did not appear in the table is a *non-singular state*. Now from a singular state, no matter how one removes the stones, the new state will be non-singular (check this). However from a non-singular state, by *suitably* removing stones, one can reach a singular state (check this). This implies if the two piles of stones were initially in a singular state, the second person wins, and conversely if the two piles of stones were initially in a non-singular state, the first person wins.

Exercises

1. Solve $4x + 3y = 2x \left[\frac{x^2 + y^2}{x^2} \right]$.
2. Solve $8[3x] - 5[2x] = 3$.
3. Solve $x^3 - [x] = 6$.
4. Find the number of real solutions of the equation $4x^2 - 40[x] + 51 = 0$.
5. Find positive x such that $\{x\} + \{\frac{1}{x}\} = 1$. Can x be rational?
6. Let $\lambda = \frac{3 + \sqrt{5}}{2}$ and n be an arbitrary integer n . Show that $[\lambda[\lambda n]] + n$ is a multiple of 3.
7. Find the number of solutions of the equation $x^2 - [x]^2 = (x - [x])^2$ within the region $[1, N]$, where N is a natural number.
8. (Canada MO 1981) Find all real x that satisfy $[x] + [2x] + [4x] + [8x] + [16x] + [32x] = 12345$.
9. (USAMO 1985) Find the number of positive integers in the range $[1, 1000]$ that can be expressed in the form $[2x] + [4x] + [6x] + [8x]$.

10. Show that the function $f(n) = [n + \sqrt{2n} + \frac{1}{2}]$ misses exactly the triangular numbers.
11. Show that $f(n) = [n + \sqrt{3n} + \frac{1}{2}]$, $n = 1, 2, \dots$ misses exactly the sequence $a_n = [\frac{n^2 + 2n}{3}]$.
12. Let $S = 1 + \frac{1}{\sqrt{3}} + \frac{1}{\sqrt{5}} + \dots + \frac{1}{\sqrt{(2n+1)^2}}$, show that $[S] = 2n$.
13. Show that $\{\sqrt{4n^2 + n}\} \leq 0.25$ for any natural number n .
14. Find all n such that $[\sqrt{n}] | n$.
15. For any natural number, there exists an irrational number α such that $k \nmid [\alpha^n]$, $n = 1, 2, \dots$.
16. Find all x such that $[x]^2 = x\{x\}$.
17. Let α be the positive root of $x^2 - 1989x - 1 = 0$. Show that there exist infinitely many n such that $[\alpha n + 1989\alpha[\alpha n]] = 1989n + (1989^2 + 1)[\alpha n]$.
18. Show that for a natural number k , there exists an irrational number α , such that for any natural number m , $[\alpha^m] \equiv -1 \pmod{k}$.
19. Let α be the largest root of $x^3 - 3x^2 + 1 = 0$. Show that $17 \nmid [\alpha^{2004}]$.
20. Let $a_n = [\sqrt{(n-1)^2 + n^2}]$, $n = 1, 2, \dots$. Show that (i) there exist infinitely many m such that $a_{m+1} - a_m > 1$, and (ii) there exist infinitely many m such that $a_{m+1} - a_m = 1$.
21. Find the highest power of 5 that divides C_{500}^{1000} .
22. Find the number of zeros in the expansion of $2007!$.
23. (Austrian-Polish MO) Find the greatest power of 2 that divides $a_n = [(3 + \sqrt{11})^{2n+1}]$.
24. (Romanian MO 2003) Find all positive integers n such that $\left[\frac{n^3 + 8n^2 + 1}{3n} \right]$ is a prime number.
25. Let p and q be relatively prime positive integers, then
$$\left[\frac{p}{q} \right] + \left[\frac{2p}{q} \right] + \left[\frac{3p}{q} \right] + \dots + \left[\frac{(q-1)p}{q} \right] = \frac{(p-1)(q-1)}{2}.$$
26. $\alpha = \frac{\sqrt{5}+1}{2}$, $\beta = \alpha^2 = \frac{\sqrt{5}+3}{2}$, show that $[\alpha[\beta n]] = [\alpha n] + [\beta n]$.
27. Let $\gamma = \frac{\sqrt{5}-1}{2}$, show that $[\gamma[\gamma n] + \gamma] + [\gamma(n+1)] = n$.
28. Let $G(0) = 0$, $G(n) = n - G(G(n-1))$, $n = 1, 2, \dots$. Find the general term of $G(n)$.

May 2012, Revised November 2022

Solution Outlines

1. Clearly $x \neq 0$, reduce the equation to $1 + \frac{3y}{2x} = \left\lceil \frac{y^2}{x^2} \right\rceil$ and set $\frac{3y}{2x} = t$ an integer, get two inequalities $1 + t \leq \frac{4}{9}t^2 < 2 + t$. Solve to get $t = -1, 3$, hence $y = -\frac{2}{3}x$ or $2x$.

2. Let $[3x] = m$ and $[2x] = n$, get $\frac{m}{3} \leq x < \frac{m+1}{3}$ and $\frac{n}{2} \leq x < \frac{n+1}{2}$. The equation $8m - 5n = 3$ implies $3n < 2m + 2$ and $3n > 2m - 3$, hence $3n - 2m = -2, -1, 0, 1$. Together get $m = n = 1$, hence $\frac{1}{2} \leq x < \frac{2}{3}$.

3. Answer: x equals 2 or $\sqrt[3]{7}$.

4. Answer: 4. Writing $x = [x] + \alpha$ and from the equation get $-40 < 4x^2 - 40x + 51 \leq 0$, hence $\frac{3}{2} \leq x < \frac{7}{2}$ or $\frac{13}{2} < x \leq \frac{17}{2}$. Separate into suitable regions to get $[x]$ and then x : $\frac{\sqrt{29}}{2}, \frac{\sqrt{189}}{2}, \frac{\sqrt{229}}{2}, \frac{\sqrt{269}}{2}$.

5. The equation is equivalent to say $x + \frac{1}{x} = n$, an integer. Hence $x = \frac{-n \pm \sqrt{n^2 - 4}}{2}$. Check that $n^2 - 4 = m^2$ does not produce any x that solves the equation, hence x cannot be rational.

6. Indeed $\lambda^2 - 3\lambda + 1 = 0$, hence $\lambda = 3 - \frac{1}{\lambda}$. Let $\frac{n}{\lambda} = k + x$ with k an integer and $x \in (0, 1)$, then $n = k\lambda + x\lambda$. Reduce $\left[\lambda \left[\frac{n}{\lambda} \right] \right] + n = 3(3n - 1) - 3k + \left[\frac{1 - x}{\lambda} \right]$ and then show $0 \leq \frac{1 - x}{\lambda} < 1$, which is equivalent to $0 \leq 1 - x < \lambda$.

7. Let $x = n + \alpha, \alpha \in \{0, 1\}$. Put into the equation and simplify to get $2n\alpha = [2n\alpha + \alpha^2]$ an integer, hence may let $\alpha = \frac{1}{2n}, \frac{2}{2n}, \dots, \frac{2n-1}{2n}$. Thus there are $2n$ solutions for $n \leq x < n + 1$. So in the range $1 \leq x \leq N$ there are altogether $2(1 + 2 + \dots + (N - 1)) + 1 = N^2 - N + 1$ solutions.

8. Let $x = [x] + \alpha$ and transform the equation to $63[x] + [\alpha] + [2\alpha] + [4\alpha] + [8\alpha] + [16\alpha] + [32\alpha] = 12345$. Now $12345 = 63 \times 195 + 60$, get $[\alpha] + [2\alpha] + [4\alpha] + [8\alpha] + [16\alpha] + [32\alpha] = 60$. However $[n\alpha] < n$, hence

$[\alpha] + [2\alpha] + [4\alpha] + [8\alpha] + [16\alpha] + [32\alpha] \leq 57$, impossible.

9. Write $x = \frac{1}{2}n + \alpha$, with $\alpha \in [0, \frac{1}{2})$. Put into the equation to get

$10n + [2\alpha] + [4\alpha] + [6\alpha] + [8\alpha]$. When $\alpha \in [0, \frac{1}{8})$, get one solution $10n$; when $\alpha \in [\frac{1}{8}, \frac{1}{6})$, get one solution $10n + 1$; when $\alpha \in [\frac{1}{6}, \frac{1}{4})$, get one solution $10n + 2$; when $\alpha \in [\frac{1}{4}, \frac{1}{3})$, get one solution $10n + 4$; when $\alpha \in [\frac{1}{3}, \frac{3}{8})$, get one solution $10n + 5$; finally when $\alpha \in [\frac{3}{8}, \frac{1}{2})$, get one solution $10n + 6$. So we have 6 solutions out of every 10 numbers, i.e. altogether 600 solutions.

10. Indeed $f(n+1) - f(n) \geq 2$ implies $\sqrt{2n} + \frac{1}{2} < k \leq \sqrt{2n+2} + \frac{1}{2}$, which can be transformed to $n < \frac{1}{2}k(k-1) + \frac{1}{8} \leq n+1$. Thus $n = \frac{1}{2}k(k-1)$. From $f(n)$ to $f(n+1)$, the number $n+k = \frac{1}{2}k(k+1)$ is omitted.

11. $\sqrt{3n} + \frac{1}{2} < k \leq \sqrt{3(n+1)} + \frac{1}{2}$ to get $n+k = \left\lfloor \frac{k^2 + 2k}{3} \right\rfloor$.

12. Check that for $k \geq 2$, $\sqrt{k+2} - \sqrt{k} < \frac{1}{\sqrt{k}} < \sqrt{k} - \sqrt{k-2}$. Summing the inequalities for $k = 3, 5, \dots, (2n+1)^2$, and by adding 1, get $\sqrt{(2n+1)^2 + 2} - \sqrt{3} + 1 < S < \sqrt{(2n+1)^2} = 2n+1$. On the other hand, $2n = \sqrt{(2n+1)^2} - 2 + 1 < \sqrt{(2n+1)^2} - \sqrt{3} + 1$, get $2n < S < 2n+1$, thus $[S] = 2n$.

13. Show that $[\sqrt{4n^2 + n}] = 2n$ and $\sqrt{4n^2 + n} < 2n + 0.25$.

14. Let $n = q^2 + r, 0 \leq r < 2q$, then $[\sqrt{n}] = q$, hence the condition implies $q \mid r$, or $r = 0, q, 2q$. Thus $n = q^2, q^2 + q, q^2 + 2q$.

15. Let $\alpha = \sqrt{k(k+1)} + k$, then $[\alpha^n] = (\sqrt{k(k+1)} + k)^n - (\sqrt{k(k+1)} - k)^n$.

16. The equation is equivalent to $\{x\}^2 + [x]\{x\} - [x]^2$. Thus $\{x\} = \frac{\sqrt{5}-1}{2}[x]$, get

$$\{x\} = \frac{\sqrt{5}-1}{2}, [x] = 1, x > 0. \text{ Hence } x = \sqrt{5} + 1.$$

17. Indeed $1989 = \alpha - \frac{1}{\alpha}$, hence $\alpha > 1$. Now

$$\begin{aligned} & [\alpha n + 1989\alpha[\alpha n]] - 1989n - (1989^2 + 1)[\alpha n] \\ &= [\alpha n + (\alpha - \frac{1}{\alpha})\alpha[\alpha n] - (\alpha - \frac{1}{\alpha})n - (((\alpha - \frac{1}{\alpha})^2 + 1)[\alpha n]] \\ &= [\frac{n}{\alpha} - \frac{1}{\alpha^2}[\alpha n]] = \left[\frac{\{\alpha n\}}{\alpha^2} \right] = 0. \end{aligned}$$

18. Take $\alpha = k + \sqrt{k(k-1)}$, then $[\alpha^m] + 1 = (k + \sqrt{k(k-1)})^m + (k - \sqrt{k(k-1)})^m$. By expanding and canceling out, get a multiple of k .

19. One checks that $2\sqrt{2} < \alpha < 3$. There is another root $\beta \in (0, 1)$ and a negative root γ . As $\alpha + \beta + \gamma = 3$, hence $1 > \beta + \gamma = 3 - \alpha > 0$. Thus $\beta > -\gamma = |\gamma|$ and $\beta^n + \gamma^n > 0$. Moreover $\beta + \gamma < 1$, and when $n > 1$,

$\beta^n + \gamma^n \leq \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) - \alpha^2 = 9 - \alpha^2 < 9 - (2\sqrt{2})^2 = 1$. Hence $\alpha^n + \beta^n + \gamma^n - 1 < \alpha^n < \alpha^n + \beta^n + \gamma^n$. Let $u_n = \alpha^n + \beta^n + \gamma^n$, we have the recurrence relation $u_{n+3} = 3u_{n+2} - u_n$ and the initial conditions $u_0 = u_1 = 3, u_2 = 9$, we see that u_n are integers and $[\alpha^n] = u_n - 1$. One also checks that $u_{n+16} \equiv u_n \pmod{17}$, hence $u_{2004} \equiv u_4 \equiv 1 \pmod{17}$, get $[\alpha^{2004}] = u_{2004} - 1$ is divisible by 17. (Indeed $\alpha \approx 2.879, \beta \approx 0.653, \gamma \approx -0.532$.)

20. Check that $\sqrt{n^2 + (n+1)^2} - \sqrt{(n-1)^2 + n^2} > 1$, so $a_{m+1} - a_m \geq 1$, one of (i) and (ii) is true. If (i) false, then for m large enough, must have $a_{m+k} - a_m = k$, $\sqrt{(m+k-1)^2 + (m+k)^2} - \sqrt{m^2 + (m+1)^2} \leq k$ for all natural numbers k . Fix m and let k goes to infinity to get a contradiction ($\sqrt{2} < 1$). If (ii) false, then for m large enough, $a_{m+1} - a_m \geq 2$, or $a_{m+k} - a_m \geq 2k$. Fix m and let k goes to infinity to get $\sqrt{2} > 2$!

$$21. \sum_k \left[\frac{1000}{5^k} \right] - 2 \sum_k \left[\frac{500}{5^k} \right] = 1.$$

$$22. \sum_{k=1}^{\infty} \left[\frac{2007}{5^k} \right] = 510, \text{ hence } 10^{510} \text{ divides } 2007!.$$

23. Let $(3 + \sqrt{11})^{2n+1} = u_n + iv_n$. Then $[(3 + \sqrt{11})^{2n+1}] = (3 + \sqrt{11})^{2n+1} + (3 - \sqrt{11})^{2n+1} = 2u_n$. One can show the recurrence relations $u_{n+1} = 20u_n + 66v_n$ and $v_{n+1} = 6u_n + 20v_n$. As $u_0 = 3$ and $v_0 = 1$, one shows by induction 2^{n+1} is the highest power of 2 dividing a_n .

24. Try $n = 3k, 3k + 1$ and $3k + 2$ to show that $n = 3$ is the only option.
25. Consider the rectangle $O(0,0), A(p,0), B(p,q), C(0,q)$, the number of integral points (points with integer coordinates) inside the rectangle is $(p-1)(q-1)$. The diagonal $y = \frac{p}{q}x$ contains no integral points (except the endpoints), hence the number of integral points above and below the diagonal will be the same $(= \frac{(p-1)(q-1)}{2})$. On the other hand, if k is an integer less than q , then the number of integral points below the diagonal on the vertical line $x = k$ is $\left\lfloor \frac{kp}{q} \right\rfloor$, the diagonal and $x = k$ meet at $(k, \frac{kp}{q})$, hence the result. (This result is a step in proving the law of quadratic reciprocity.)
26. $\beta = \alpha + 1$, get $\{\alpha n\} = \{\beta n\}$ Thus
 $\alpha[\beta n] - [\beta n] = (\alpha - 1)(\beta n - \{\beta n\}) = \alpha n - (\alpha - 1)\{\alpha n\} = [\alpha n] + (2 - \alpha)\{\alpha n\}$, which lies between $[\alpha n]$ and $[\alpha n] + 1$, OK.
27. Let $\{\gamma n\} = \varepsilon$, show that $[\gamma[\gamma n] + \gamma] + [\gamma(n+1)] = n + [\frac{1}{\gamma}(1 - \gamma - \varepsilon)] + [\varepsilon + \gamma]$. If $2 > \gamma + \varepsilon > 1$, the last two terms are -1 and 1 . If $\gamma + \varepsilon < 1$, then $0 + 0 = 0$. Note $\gamma + \varepsilon \neq 1$, otherwise $\gamma(n+1) = [n\gamma] + 1$, contradicting the irrationality of γ .
28. $G(n) = [\gamma(n+1)]$. Use the above question and the Beatty's theorem.

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